Shauryam Bajpai

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Experience

NetProphets Cyberworks Private Limited | Software Engineering Intern May 2026 – July 2026

- Built and shipped AI-powered backend services in Python/FastAPI, containerized with Docker, delivering 15 features/endpoints into production within an Agile sprint cycle.
- Integrated LLM inference and vector database calls into the core application, working across the Next.js/TypeScript frontend and Python backend to ship end-to-end features.
- Designed structured logging and input validation around LLM and vector database calls, cutting failure triage time by 28% and eliminating silent timeout and retry failures that had previously gone untracked in production.

Projects

LocalMind: Private AI Document Intelligence System 2026

Tech Stack: Python, FastAPI, Next.js 14, Typescript, Qdrant(Vector DB), Ollama, Sentence-Transformers, Pytesseract(OCR), Tailwind CSS, Docker.

Developed a production grade, privacy-first fully local RAG (Retrieval Augmented Generation) system that enables private AI powered document querying and chat with LLM with zero data leaving the machine – 100% offline.

- Engineered a complete RAG pipeline from scratch implementing word-level chunking, sentence-transformer embeddings, and Qdrant vector search enabling accurate, multi-document retrieval with OCR support for scanned PDFs.
- Built a streaming FastAPI backend with local LLM inference via Ollama, persistent chat history, and a Next.js frontend with real-time token streaming and markdown rendering all completely containerized via Docker with zero cloud dependency.
- Optimized embedding and retrieval latency through batched inference and Qdrant index tuning, reducing average query response time by 42% and enabling sub-2-second retrieval across 10,000+ document chunks.

PeerLink Storage: High-Performance Distributed File Storage System 2026

Tech Stack: Go, TCP/IP, Content-Addressable Storage (CAS), AES Encryption, Docker, gRPC, Protobuf.

Architected a decentralized storage provider in Go leveraging a peer-to-peer architecture to coordinate secure, distributed file management across a multi-node cluster.

- Built a custom P2P transport protocol and handshake over raw TCP, leveraging Goroutines and Channels for non-blocking, low-latency file streaming.
- Implemented AES encryption and automated replication for data privacy and high availability, alongside a centralized metadata store for efficient file discovery and retrieval across nodes.
- Wrote unit and integration tests covering the P2P handshake, chunked file transfer, and encryption/decryption paths, catching race conditions in concurrent Goroutine access before they surfaced in multi-node runs.

SplitLedger: Concurrent Expense and Debt Settlement Platform 2026

Tech Stack: Java, Spring Boot, Spring Security, React, TypeScript, PostgreSQL, JDBC, Flyway, JWT, JUnit, Testcontainers, Docker.

Built a full-stack expense-management platform that calculates shared balances, minimizes repayments and processes concurrent financial updates through secure, auditable workflows.

- Architected 17 REST APIs and a normalized 9-table PostgreSQL schema supporting multi-user groups, multi-currency balances, JWT authentication and 3 configurable splitting strategies: equal, exact and percentage.
- Developed an O(n log n) debt-simplification algorithm using priority queues, reducing potentially $O(n^2)$ pairwise obligations to at most n−1 settlement transactions for a group of n users.
- Implemented row-level locking, transactional rollbacks, idempotency keys and immutable audit logs to prevent duplicate or inconsistent financial

Technical Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript, Go

Frameworks & Libraries: React, Node.js, FastAPI, REST APIs, Express, Next.js, gRPC, Tailwind CSS, WebSockets

AI/ML: RAG, Vector Databases, Embeddings, LLMs, Sentence-Transformers, Ollama, OpenAI API

Databases: Oracle SQL, MySQL, MongoDB, PostgreSQL, Qdrant, Redis

Developer Tools: Git, GitHub, Docker, Postman, Chart.js, VS Code, Eclipse, Tesseract OCR

Competitive Skills: Solved 400+ Leetcode problems

Certifications

- **Building AI Solutions Using Advanced Algorithms and Open Source Frameworks - IBM (Skillsbuild)**
- **Data Structures - UC San Diego (Coursera)**
- **Java and Object-Oriented Programming – University of Pennsylvania (Coursera)**

Education

Manipal Institute of Technology Bengaluru, KA
Bachelor of Technology in Computer Science *Aug 2023 - Present*

City Montessori School, Mahanagar Lucknow, UP
Higher Secondary Education (Class XII) *April 2022 - June 2023*